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Introduction to Movement Science Theory

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What do you know about Movement Science? What do you think it is?

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The Four Areas

- Normal Motor Control Model
- Motor Skill Acquisition
- Variables affecting Practice and Learning (VAPAL)
- Dyscontrol and Recovery of Function

Tools Available to the Therapist

- VAPAL
- Applying resistance or assistance
- Modalities
- Inhibitory pressure
- Traction/compression
- Vibration

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What is Motor Control?

Motor Control – the ability to regulate or direct the mechanisms essential to movement

…helps us answer



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What is Motor Control?

Motor Control – the ability to regulate or direct the mechanisms essential to movement

...helps us answer

- How does the central nervous system (CNS) organize the many individual muscles and joints into coordinated functional movements?
- How is sensory information from the environment and the body used to select and control movement?
- How do our perceptions of ourselves, the tasks we perform, and the environment in which we are moving influence our movement behaviour?
- What is the best way to study movement, and how can movement problems be quantified in patients with motor control problems?

(Shumway-Cook, 3)

Motor Control

Movement emerges from the interaction of 3 factors:

1. Individual
2. Task
3. Environment

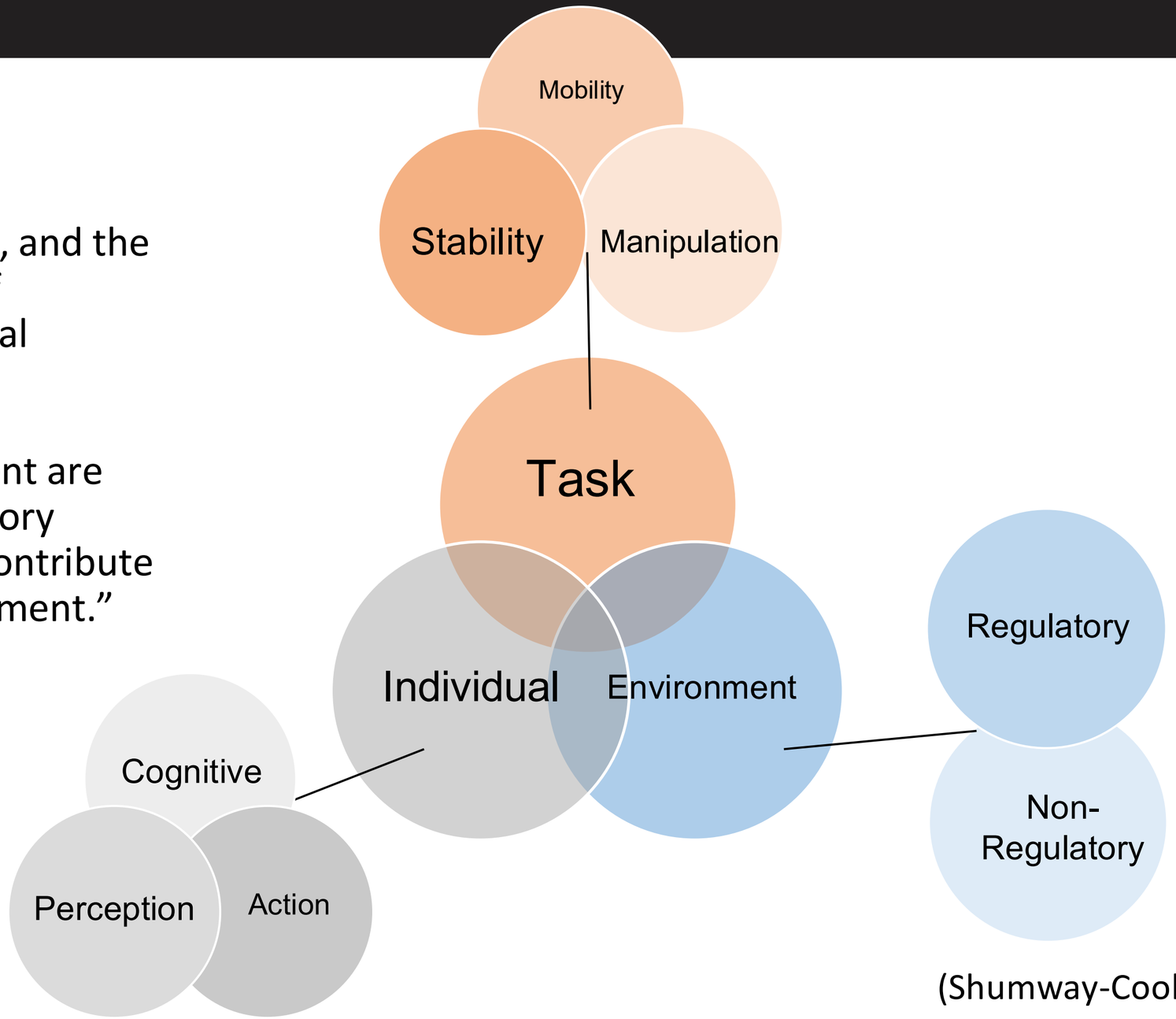
Movement is organized around both task and environmental demands. The individual generates movement to meet the demands of the task being performed within a specific environment.

Motor Control includes perception and action systems that are organized to achieve specific goals or intents. This the study of motor control must include the study of cognitive processes as they relate to perception and action

(Shumway-Cook, 3)

Motor Control

"Factors within the individual, the task, and the environment affect the organization of movement. Factors within the individual include the interaction of perception, cognition, and action (motor) systems. Environmental constraints on movement are divided into regulatory and nonregulatory factors. Finally, attributes of the task contribute to the organization of functional movement."



(Shumway-Cook, 5)

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You are kicking a penalty kick and the keeper moved left, so instead of kicking straight you kicked the ball to the right



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0 Motor Equivalence

0 Movement Equivalence

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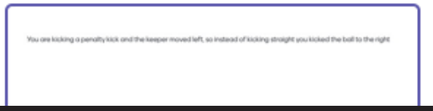
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Motor Control

Human behavior is goal driven:

- Accomplish intentional activities
- Everything that you do is done to accomplish intentional activities/actions
- If the intention is not there, you look confused

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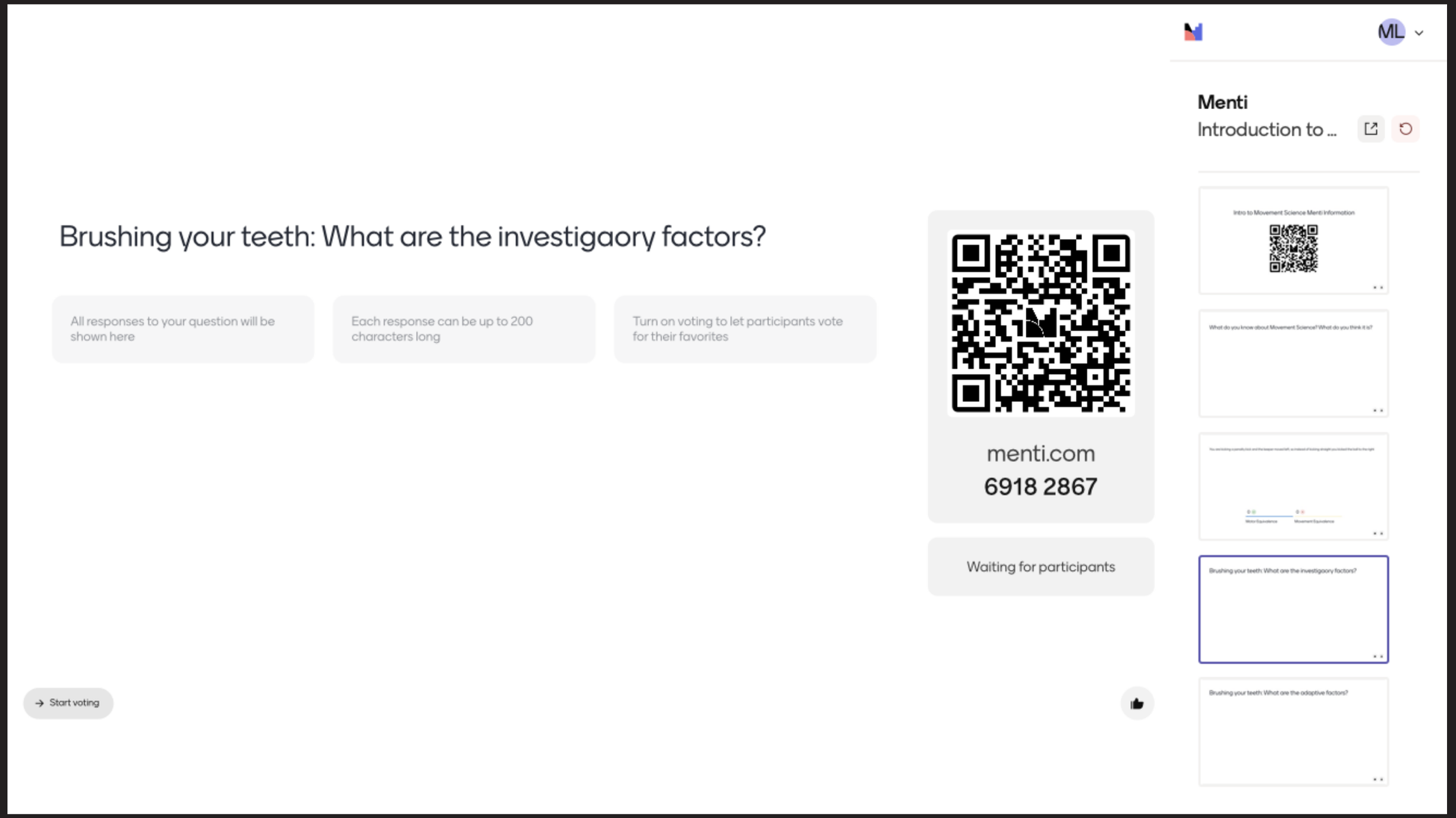
Motor Control

1. **Investigatory**
   - Pick up information from the environment; orient, focus, adjust our sensory analyzers to gather information
   - We use our sensory analyzers to pick out very important information in the environment – like gathering supplies
   - Example: when going to make dinner, you go into the kitchen, you gather the supplies that you need and use sensory analyzers to look for and find what you need
   - Ex – when crossing the street (investigatory), feel the wind, listen for cars, etc

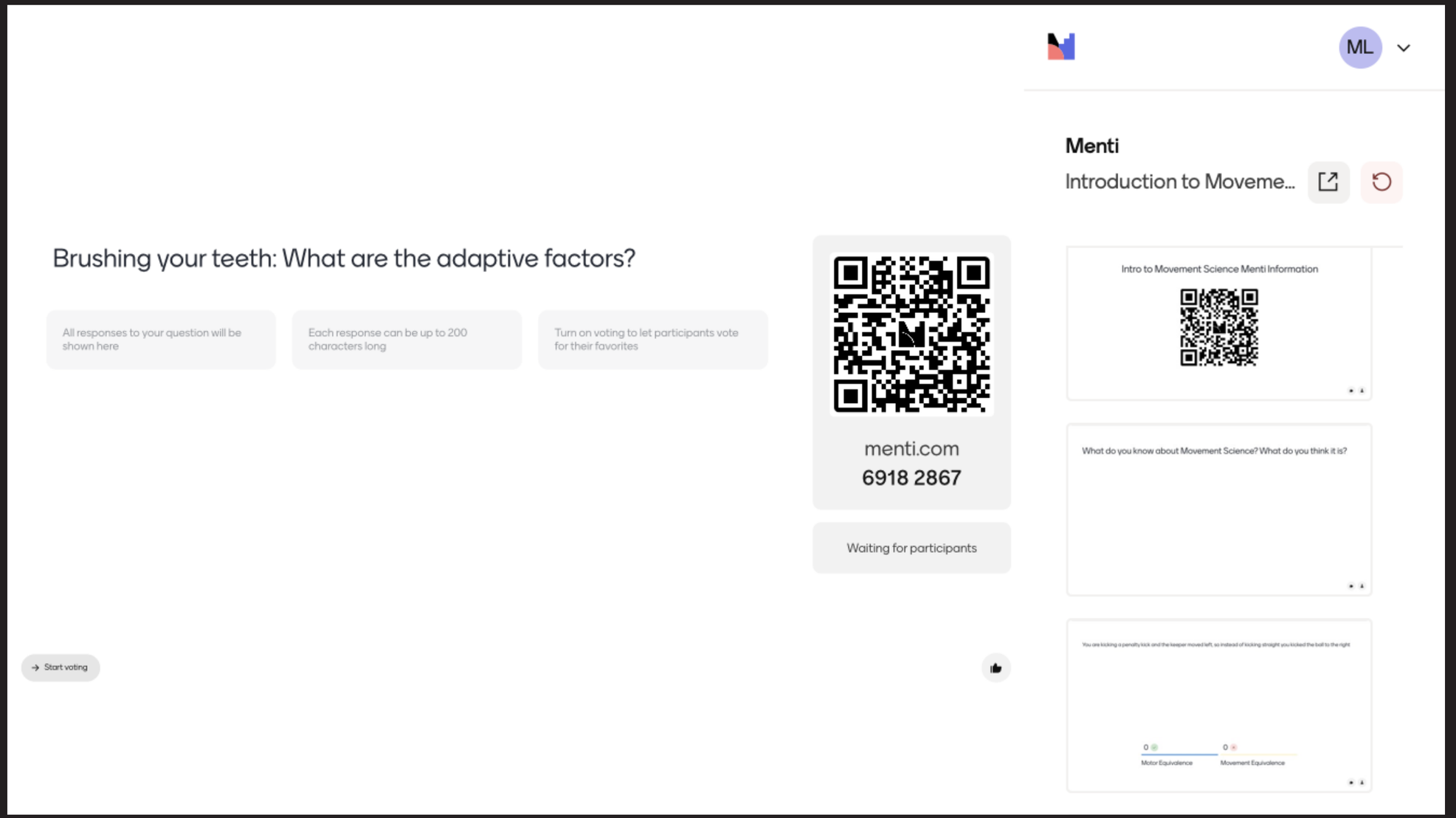
2. **Adaptive**
   - Interact directly with the physical or social environment
   - Once you have information, adaptive behavior kicks in because this is the behavior that interacts with the physical or social environment
   - Ex - The actual act of crossing the street or the act of avoiding things in the street

(Shumway-Cook, 3)

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Using Motor Control to Analyze Behavior

Action
(what)

1. Action (Behavior or Task)

- That is the actual outside/end of what you did
- Action: change in state from initial to final conditions (not how the change has occurred)
- Not how the change occurred, but it is what occurred
- OT and PT functional goals fit very nicely in an action because they say what is done, can be successful

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Using Motor Control to Analyze Behavior

Action
(what)

Movement
(how)

2. Movement (Biomechanical)

- Strategies, patterns, path
- Form (topology) and detailed sequence of components (joint motion, muscles, joints, torques, EMG)

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Using Motor Control to Analyze Behavior

Action
(what)

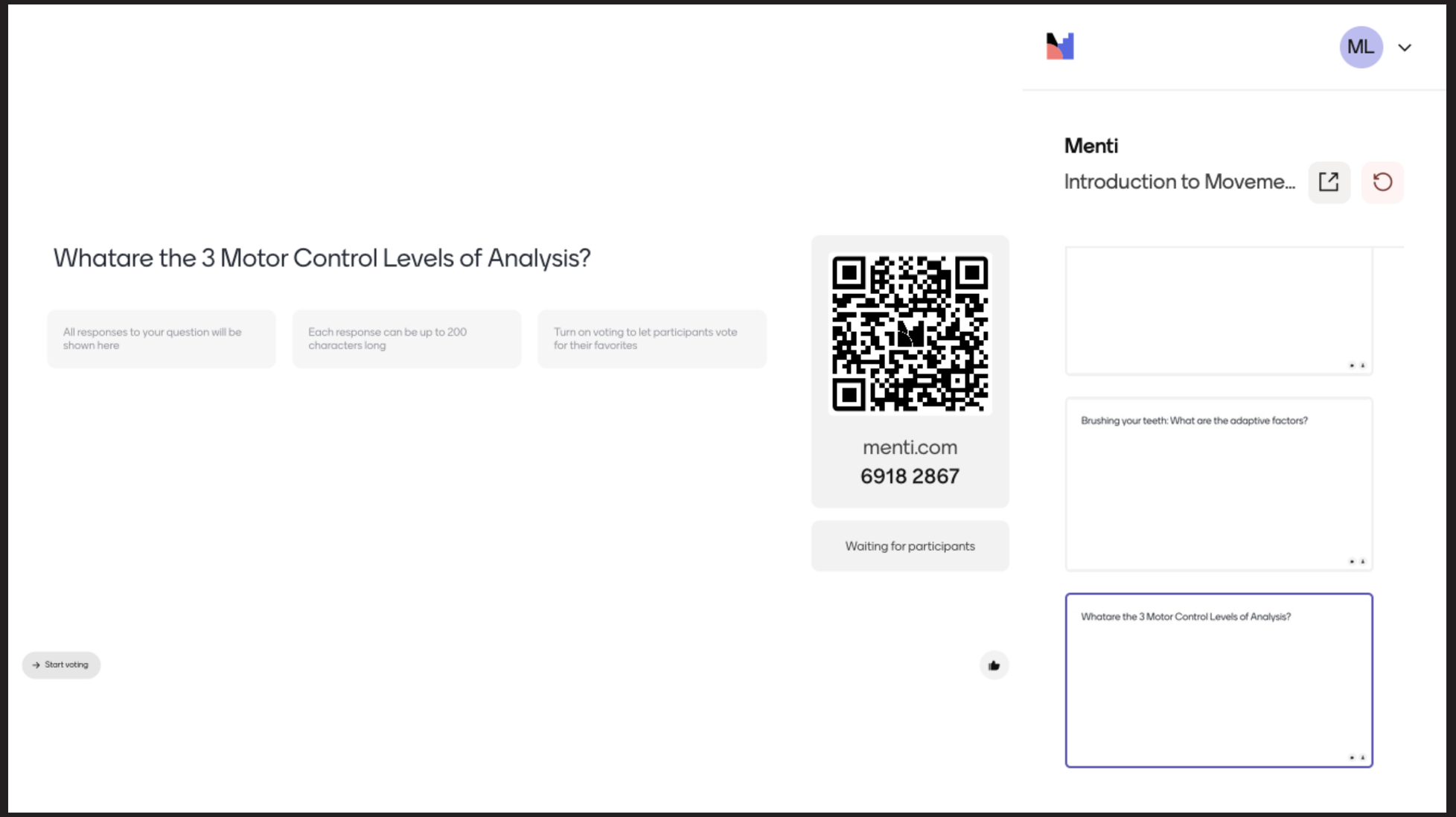
Movement
(how)

Neuromotor Processes
(sensory and motor neurons needed)

3. Neuromotor Processes

- Look at organization of the action on the basis of which brain areas are working, which tracts, whether it is at the brain or spinal cord level, which peripheral nerves are being utilized
- Organizational mechanisms in the CNS that constrain and sequence the movements
- Usually use inference to depict what is happening here because we see inside head

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Using Motor Control to Analyze Behavior – Relationship between levels is not 1:1

Action
(what)

Movement
(how)

Neuromotor Processes
(sensory and motor neurons needed)

Using Motor Control to Analyze Behavior – Relationship between levels is not 1:1

How to you write with a pen?

- Option 1: https://www.youtube.com/watch?v=9Ldwr2LLAos (start at 1:07)
- Option 2: https://www.youtube.com/watch?v=Va1InskEgsc
- Option 3: https://www.youtube.com/watch?v=mlMQw6xMn6k

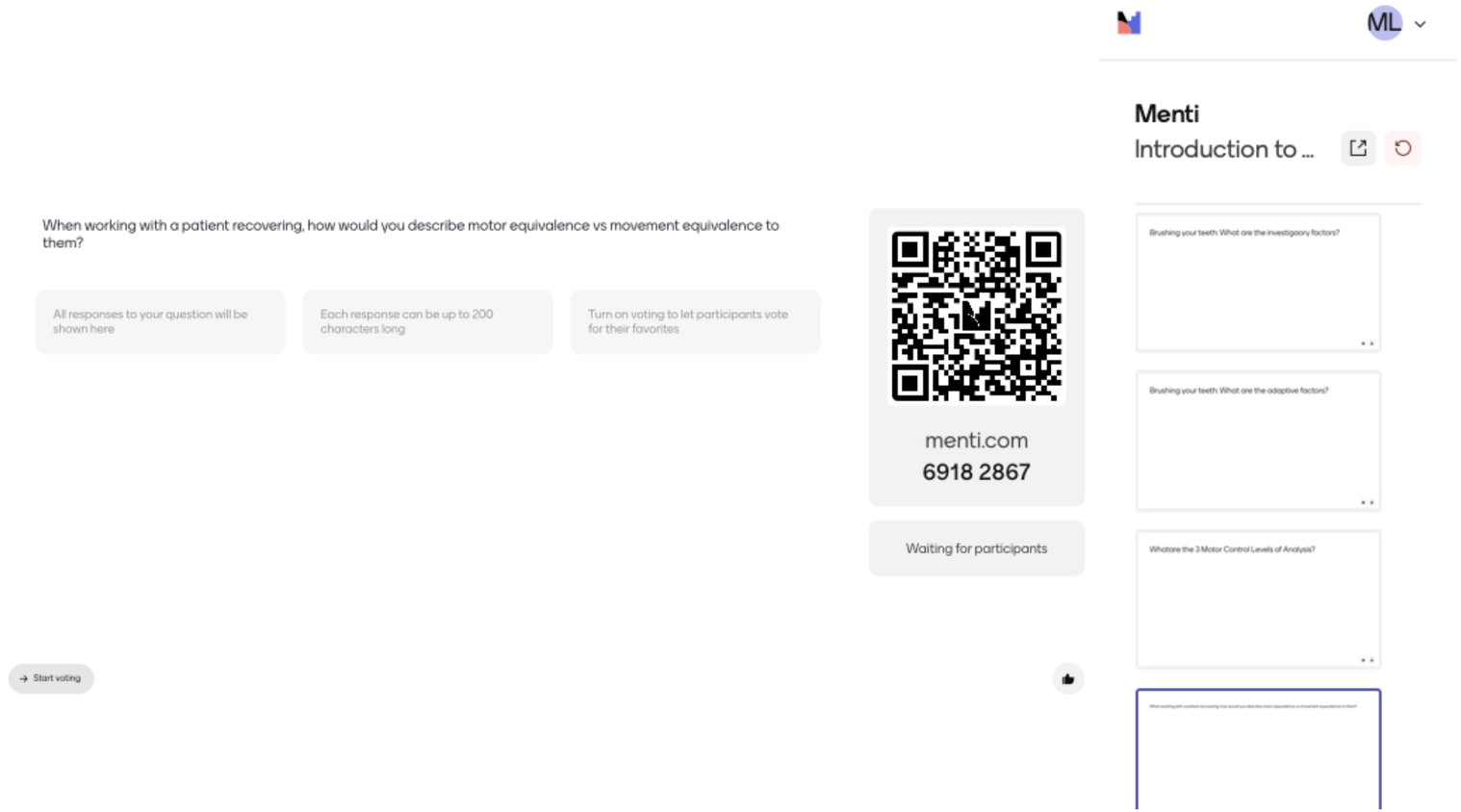
Using Motor Control to Analyze Behavior

- Action can drive the movement
- Action can drive the neuromotor processes
- Many movement patterns which can produce the same outcome
- **Movement equivalence** is changing the way you MOVE to accomplish the same task
- **Motor Equivalence** is changing the NEUROMOTOR PROCESSES you use to accomplish the same task

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Using Motor Control to Analyze Behavior

Motor Equivalence vs Movement Equivalence



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Part 2 – Skill Acquisition

Skilled Action

- Competence
- Consistency & Economy
- Effective Use of Resources
- Relative to the Individual
- Task Specific

Skilled Action









Skill Acquisition

Skill - Consistency in attaining the goal with some economy of effort

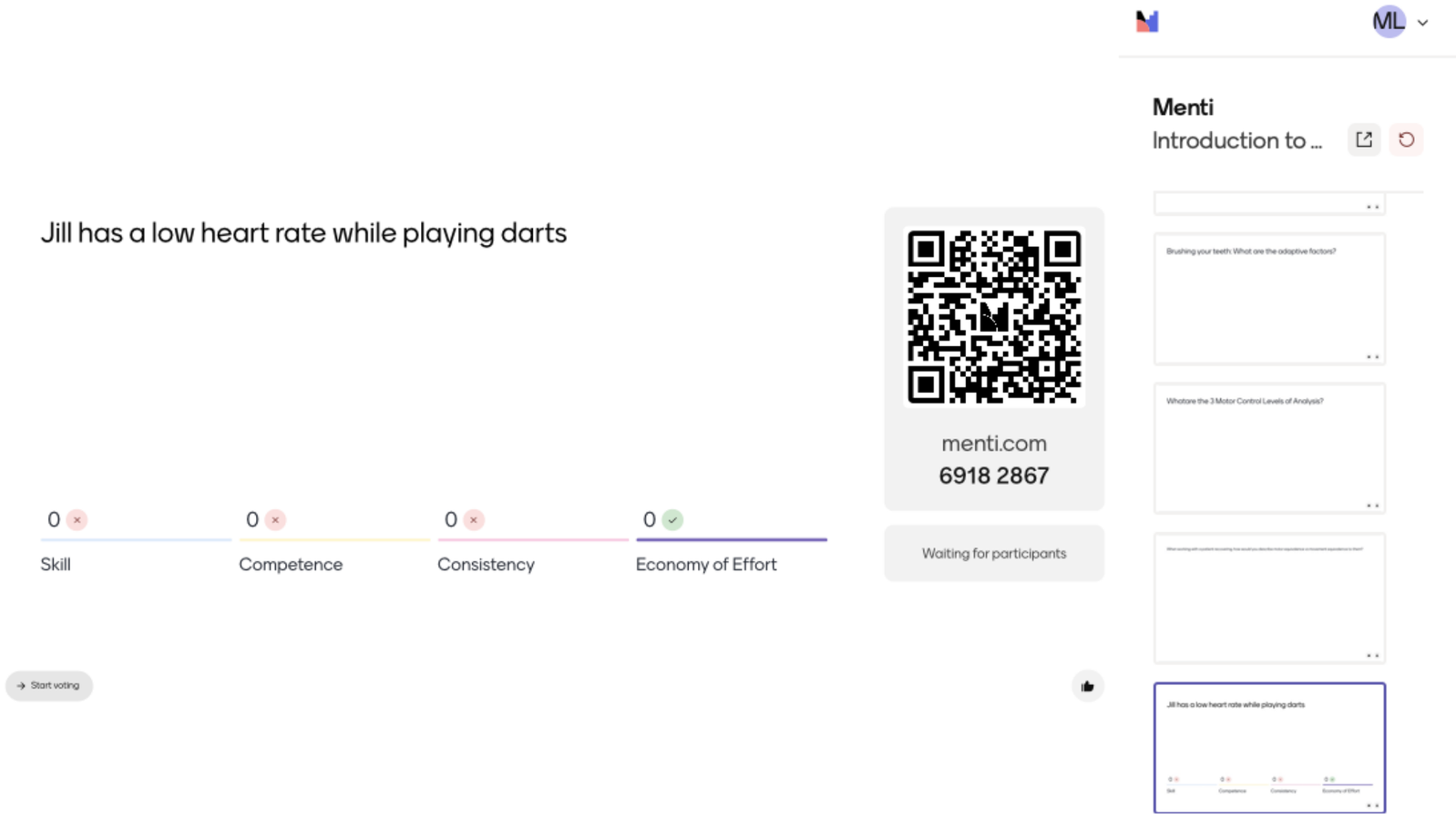
Competence – When someone has a skill

Consistency - Able to repeat something (motor skill) with small changed in standard deviation

Economy of effort - Low energy demand (O2/CO2 ratio, rate of perceived exertion, baseline and change in heart rate, time it takes to do something)

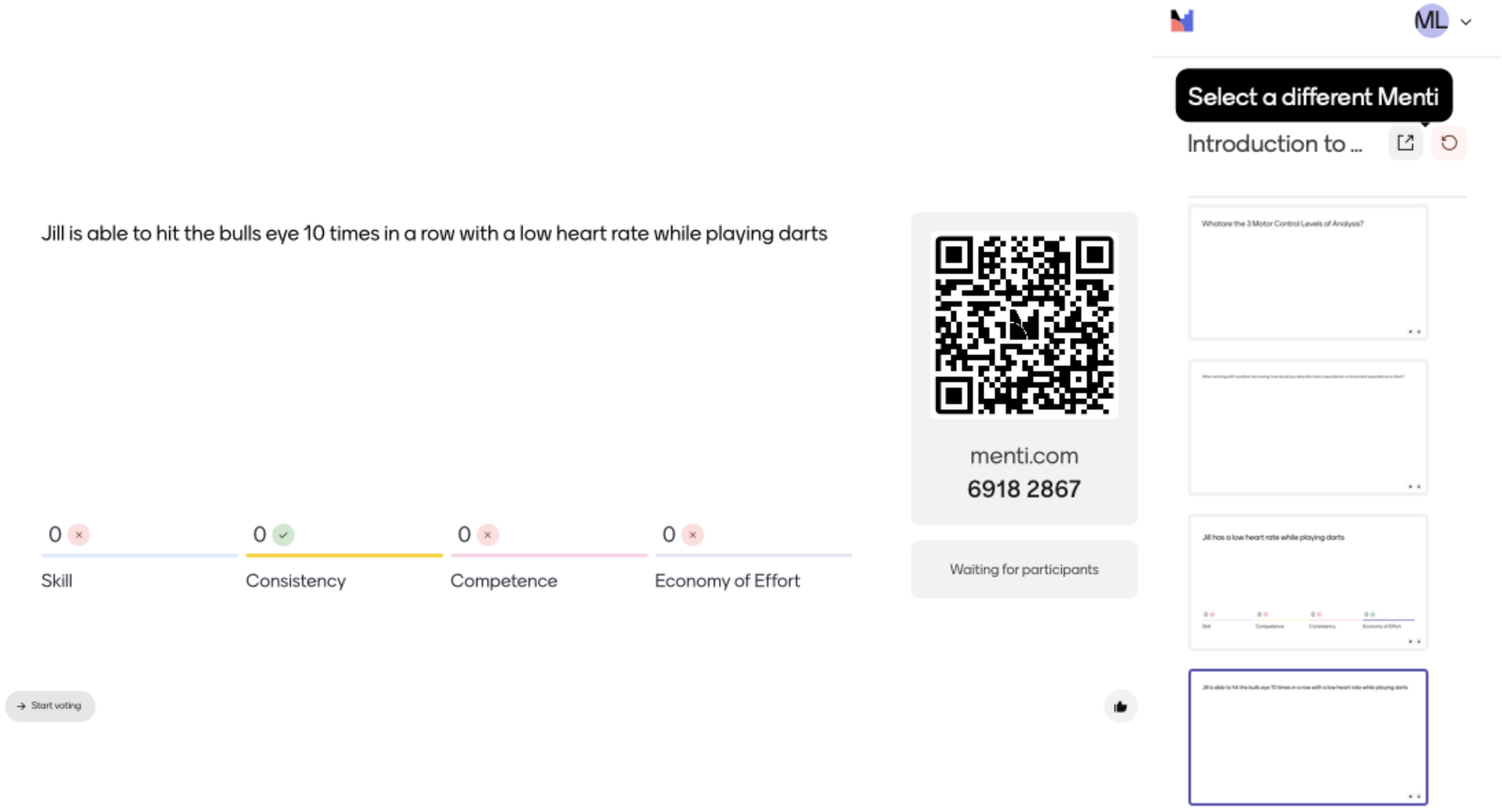
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Jill has a low heart rate while playing darts



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Jill is able to hit the bulls eye 10 times in a row with a low heart rate while playing darts



Skilled Action







Motor Theories

Some theories:

- **Motor Programming Theory**
- **Systems Theory**
- **Ecological Theory**

Current Theory used by Physical Therapists & Physiotherapist:

- **Dynamical System Theory** –
  - Movement results from many systems interacting (constraining) to yield the most stable pattern
  - Subsystems self-organize
  - Subsystems organize at different rates

“Movement is possible in the absence of reflexive action. Sensory input, while not essential in driving movement, has an important function in modulating action.” (Shumway-Cook, 11)

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Skill Acquisition:
Learning

Learning - defined as a relatively permanent change

True learning - measured when you stop doing the activity, you go away, and you come back *(retained learning)*

Motor Learning – the study of the acquisition and/or modification of movement.

While motor control focuses on understanding the control of movement already acquired, motor learning focuses on understanding the acquisition and/or modification of movement

Performance - temporary change in motor behavior seen during practice sessions

For example, Mrs. Jean J shows an improved ability to stand symmetrically (with weight evenly distributed to both legs) at the end of her daily therapy session, but when she returns to therapy the following day, she again stands with all her weight on her noninvolved leg. This suggests that while performance had improved in response to therapy, learning had not yet occurred. When on subsequent days Mrs. J demonstrates a more symmetric weight-bearing stance even as she arrives for therapy, we may suggest that learning (a permanent change in behavior) is occurring.

(Shumway-Cook, 21-22)

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Skill Acquisition:

Factors that Affect Learning

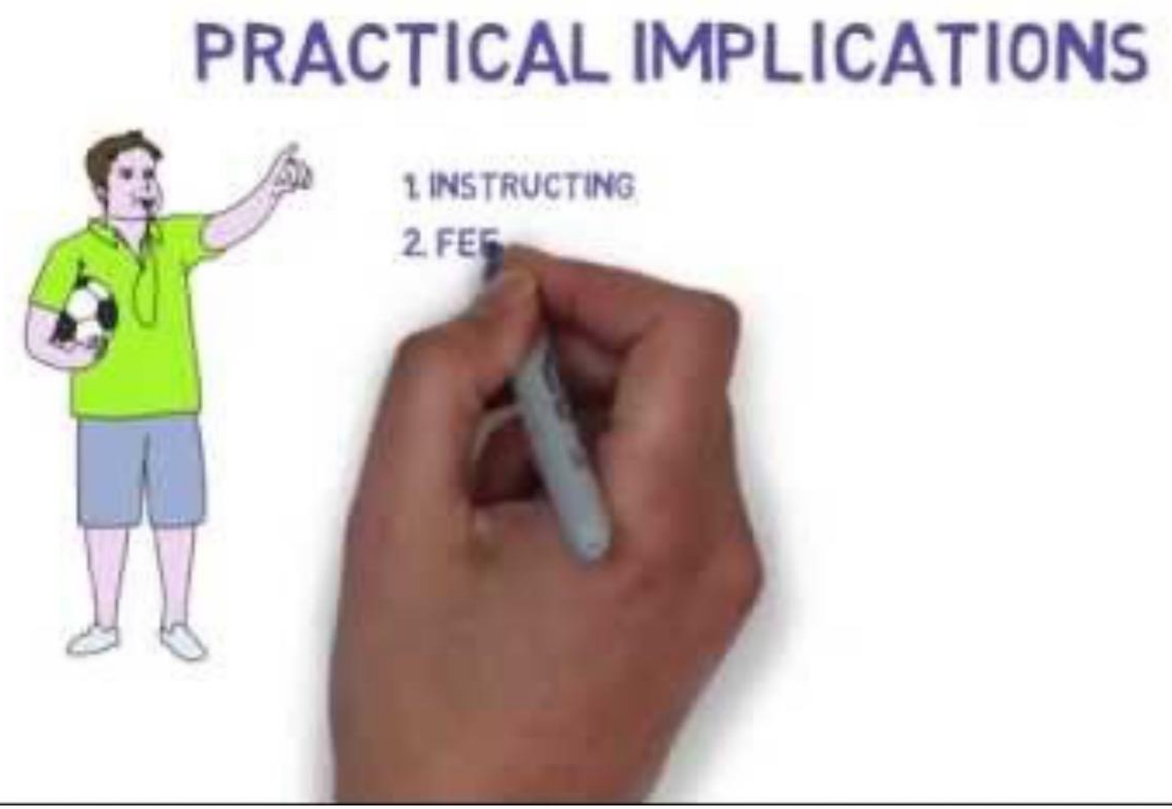
Arousal

Motivation

Attention

Memory

Does attention affect Motor Learning vs Motor Performance? YES



Improving Neuromuscular Control using Task Specific Practice

- Improving performance should be specific to what should be achieved
- Mechanisms underlying training motor control- neuroplasticity & motor learning

Implementing Neuroplasticity motor control programmes	Applying motor learning principles to motor control programmes
Meaningful activities e.g. sit to stand to get to toilet, to get food, to go out	**High Intensity Practice:** within and outside practice
High Intensity: 3 – 4 times a week, at least 2 hours per day crucial	**Feedback:** intrinsic & extrinsic feedback
High repetition: 300 reps for upper limb and 2000 steps for lower limb	**Practice conditions:** modification of practice elements to enable task to be performed
Task specificity: acquisition of skills required and not anatomical movements	**Focus of attention:** paying attention to interaction with objects rather than movements is more meaningful
Challenge: task must involve dual tasking and/or multi-tasking	**Instructions:** modelling or verbal instructions
Variation and Timing: arrange task to start from various starting position	

Other Approaches

Technique	Uses
Hydrotherapy	Improved ability to move in water
Gymnastic balls	To support patient, train balance
Proprioceptive Neuromuscular Facilitation	Promote muscle contraction
Treadmill training	Cardiorespiratory and repetitive mobility training
Balance training	Improve balance
Constrained induced movement training	Forced upper limb use
Cueing	Prompting movement
TENS	Pain Relief, muscle stimulation, spasticity relief
Neurodynamics	Promoting neural mobility
Orthoses	Promote function and support joints

Put Motor Learning Into Practice!

Let’s practice teaching someone a skill!

... Like learning to juggle



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Let's make a list of "the best" type of cues for someone to learn to juggle

All responses to your question will be shown here

Each response can be up to 200 characters long

Turn on voting to let participants vote for their favorites

Start voting

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